

Computer Modeling & Simulation Problems

The owner of car renting company, pays a penalty for each accident according to the following distribution:-

Amount of money/accident	Probability
50 L.E	0.30
100 L.E	0.30
150 L.E	0.25
200 L.E	0.15

The accidents occur with the probability of 30% (number of accidents per day is up to 6). Number of accidents per day has the following distribution:-

Number of accidents	Probability
2	0.05
3	0.15
4	0.40
5	0.25
6	0.15

Simulate 12 days. Based on your simulation what is the minimum money the owner must have to cover the penalties occurred per day. Use the following R.N:

**92 44 99 15 97 21 47 80 28 87 13 33 42 84 27 64 59 33 84 00 10 50
51 09 31 12 94 96 97 77**

Solution:

The probability accidents occur is 30% → 00 --> 29

The probability accidents don't occur is 70% → 30 --> 99

Number of accidents	Probability	
2	0.05	00—04
3	0.15	05—19
4	0.40	20—59
5	0.25	60—84
6	0.15	85—99

Amount of money/accident	Probability	
50 L.E	0.30	00—29
100 L.E	0.30	30—59
150 L.E	0.25	60—84
200 L.E	0.15	85—99

92 44 99 15 97 21 47 80 28 87 13 33 42 84 27 64 59 33 84 00 10 50
51 09 31 12 94 96 97 77

1st Day:

RN: 92 → NO ACCIDENT

2ND Day:

RN: 44 → NO ACCIDENT

3RD Day:

RN: 99 → NO ACCIDENT

4TH Day:

RN: 15 → ACCIDENT

RN: 97 → Number of accidents = 6 accidents.

→ 1th Accident:

RN: 21 → Amount of money paid= 50 L.E

→ 2ND Accident:

RN: 47 → Amount of money paid= 100 L.E

→ 3rd Accident:

RN: 80 → Amount of money paid= 150 L.E

→ 4th Accident:

RN: 28 → Amount of money paid = 50 L.E

→ 5th Accident:

RN: 87 → Amount of money paid = 200 L.E

→ 6th Accident:

RN: 13 → Amount of money paid = 50 L.E

- Then the total amount paid in this day = $50 + 100 + 150 + 50 + 200 + 50$
= 600 L.E

5th Day:

RN: 33 → NO ACCIDENT

6TH Day:

RN: 42 → NO ACCIDENT

7th Day:

RN: 84 → NO ACCIDENT

8th Day:

RN: 27 → ACCIDENT.

RN: 64 → Number of accidents = 5 accidents.

1st Accident:

RN: 59 → Amount of money paid = 100 L.E

2nd Accident:

RN: 33 → Amount of money paid = 100 L.E

3rd Accident:

RN: 84 → Amount of money paid = 150 L.E

4th Accident:

RN: 00 → Amount of money paid = 50 L.E

5th Accident:

RN: 10 → Amount of money paid = 50 L.E

- Then the total amount paid in this day = $100 + 100 + 150 + 50 + 50 =$
450 L.E

9TH Day:

RN: 50 → NO ACCIDENT.

10TH Day:

RN: 51 → NO ACCIDENT.

11TH Day:

RN: 09 → ACCIDENT.

RN: 31 → Number of accidents = 4 accidents

→1st Accident:

RN: 12 → Amount of money paid= 50 L.E

→2nd Accident:

RN: 94 → Amount of money paid= 200 L.E

→3rd Accident:

RN: 96 → Amount of money paid= 200 L.E

→4th Accident:

RN: 97 → Amount of money paid= 200 L.E

- Then the total amount paid in this day = $50 + 200 + 200 + 200 =$ 650 L.E

12th Day:

RN: 77 → NO ACCIDENT

- Based on the above simulation :- The minimum money the owner must have to cover the penalties occurred per day (The maximum paid money in all days)

= 650 L.E